

CO1 _ AT2

CSA0612 - Design and Analysis
of Algorithms.

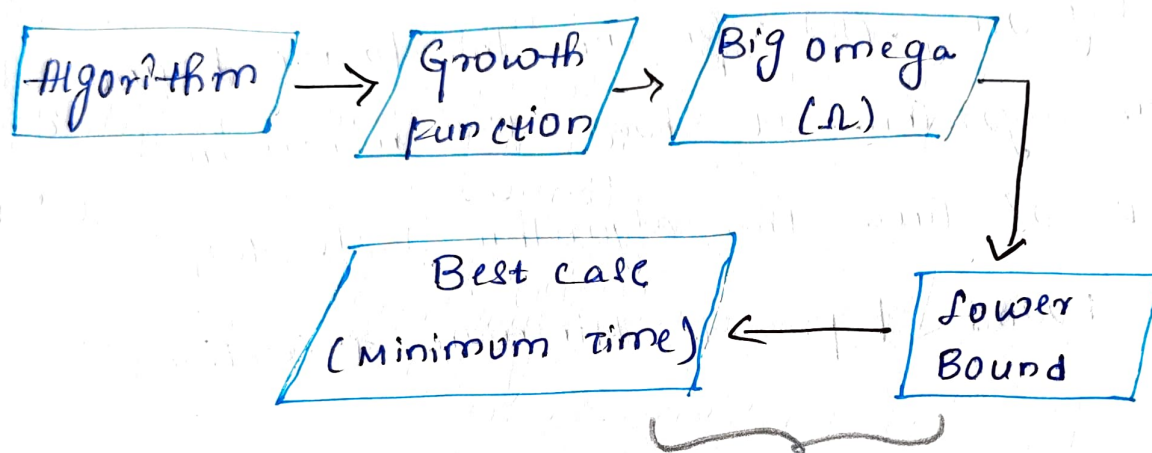
Name: P. Hemika

Reg No: 192424259.

Q 37) Consider the concept map:

Algorithm $\rightarrow$ Growth function $\rightarrow$ Big omega
 $\rightarrow$ Lower Bound.

Concept Map



Performance Guarantee

(At least $\Omega(f(n))$ time)

Algorithm: A problem solving procedure

Growth function: Describes how time/space grows with input size

Big. omega: Provides a lower bound on the growth function.

Lower Bound: Minimum time taken by the algorithm in any case

Best case: Best possible input scenario with minimum execution time

Explanation:

A growth function describes how the running time of an algorithm changes as the input size (n) increases.

Big Omega (Ω) represents the lower bound of an algorithm. It shows the minimum amount of time the algorithm will take.

Lower Bound Represents Minimum Execution time.

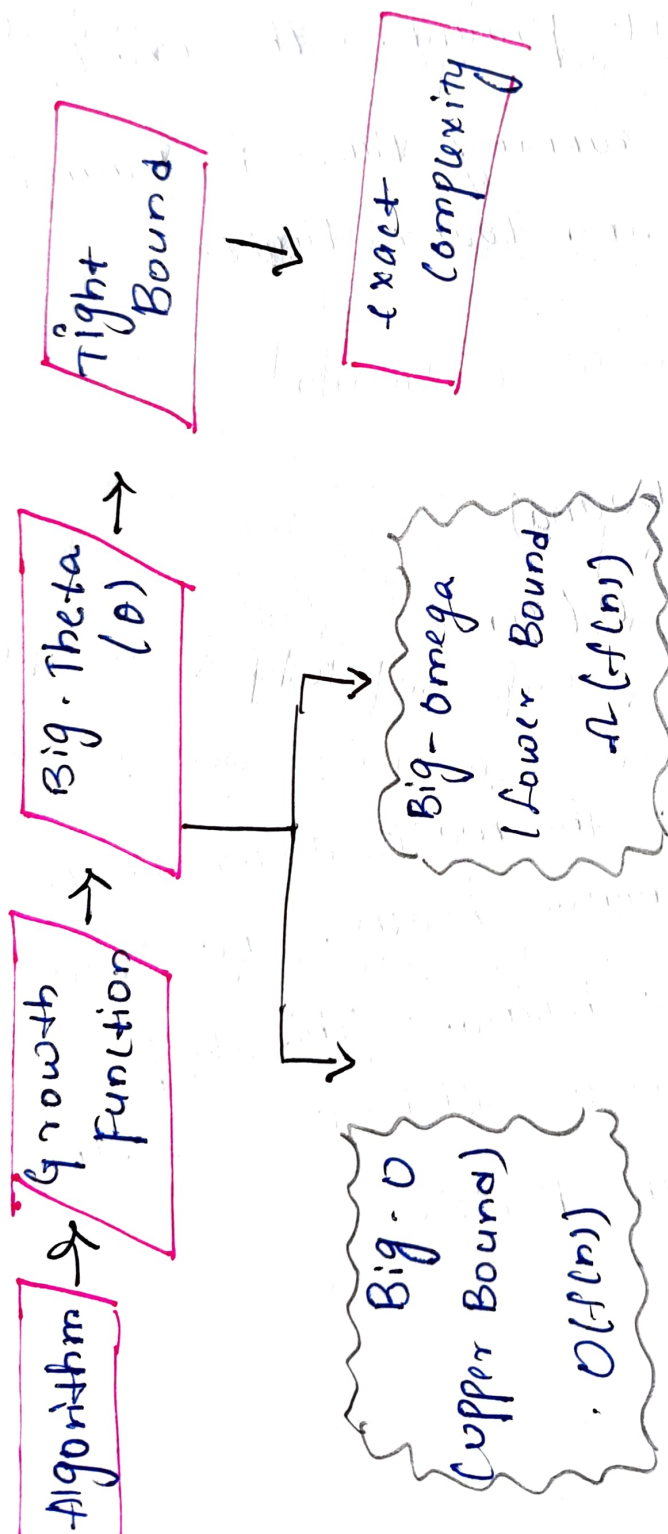
Interpretation:

Best Case Analysis provides the minimum performance limit of an algorithm. When combined with average-case and worst case analyses, it gives a complete understanding of algorithm's efficiency.

Consider the Concept Map:

Algorithm $\rightarrow$ Growth function $\rightarrow$ Big. Theta $\rightarrow$
Tight Bound $\rightarrow$ Exact

Concept Map



Big-O (Upper Bound) - algorithm grows no faster than $O(f(n))$

Big-Omega (Lower Bound) - algorithm grows no slower than $\Omega(f(n))$

Big-Theta (Tight Bound) - exact growth rate, both upper bound

& lower bound.

Explanation:

A growth function shows how an algorithm's running time increases with input size. Big theta (Θ) represents the tight bound of an algorithm. It provides the exact time, meaning the running time is bounded by both an upper and lower limit.

Link:

Big O (Upper Bound)

Big - Omega (Lower Bound)

Big - Theta ($\Theta(f(n)) \cap O(f(n))$)

Interpretation:

Big - Theta provides the most accurate measure of algorithm performance because it considers both upper and lower bounds. This helps developers choose the most efficient algorithm and make better design decisions.